Morphological Classification of Galaxies in HST and JWST Images Using Classical DIP

ECE 501 Digital Image Processing — Mid-Semester Evaluation Report

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***Abstract*—The objective of this report is to highlight the mid-term evaluation of an end-to-end pipeline for the morphology classification of galaxies through classical DIP techniques on the data obtained from the Hubble Space Telescope and the James Webb Space Telescope. In order to demonstrate the mathematics and physics involved, the following non-learning methods are followed by the pipeline: no neural networks, no trained classifiers, and no astrophysics. The first four-weeks' deliverables, including classic DIP deck exercises, sampling, aliasing, quantization, affine transformation, MAST FITS ingestion, Wiener Deconvolution, bilateral denoising, ASINH stretch, adaptive segmentation, star-galaxy separation, and CAS feature extraction, have been achieved and mathematically verified against Galaxy Zoo 2.**

***Index Terms*—Digital Image Processing, HST, JWST, Deconvolution, Adaptive Segmentation, CAS Morphology.**

I. Introduction & Problem Scoping

The HST imagery (optical/UV) and JWST imagery (NIR) are face-to-face. range (HDR) imaging and (2) low signal intensity.The two severe digital image processing challenges are: (1) **High dynamic range (HDR) imaging** and (2) **low signal intensity**. The highest values of this range (> 90 dB) are occupied by the galaxy centers. There is no possibility of a linear scale, and galactic tidal arms are lost in noise; (2) **Blurring and diffraction caused by an Optical Point Spread Function**. (3) **Poisson noise, read out noise, and cosmic rays**; (4) **Segmentation not detected**, and crowded field of galaxies, with interacting galaxies having overlapping halos.

As opposed to the ML Fundamental Assumptions & Limitations: The project that is currently being undertaken is solely Without machine learning or any other method, a deterministic classical DIP. Any other type of neural network (no CNNs, SVMs, embeddings) or any astrophysics (without cosmological red shift or stellar population). The problem has been stated to be purely deterministic 2D DIP pipeline involving steps of restoration, seg- mentation, CAS The characteristics of the image, and morphological classification. All tasks for Weeks 1-4 have been accomplished.

Related Work: The classification of galaxy morphology was done in the following way: Related Work: Galaxy morphology classification according to the classical scheme, they have been dependent on the use of CAS.The project is to identify early warning indicators using criteria and Gini-M20 coefficients. From the late type galaxies both with and without the absence of any supervised methods of learning. The one project that was working on a dealing had to do with the pressure. Galaxy Zoo, which has been providing classification of galaxies with since 2005. Its classifications as the reference for the benchmarking of both classical and machine-learned classifiers. Richardson-Lucy, Special non-linear stretches, deconvolution Wiener/ MMSE. Image restoration is the foundation for the used in astronomy. Process used in this project.

II. Foundational ECE 501 Deck Exercises

The eight basic assignments of Digital Image Processing have been accomplished and checked:

- **A1-A2. The concepts of spatial sampling and interpolation are discussed.** Subsampling of an image with size 512×512 with factors $1/2, 1/4, 1/8$ (`cv2.INTER_NEAREST`). Small source objects will be invisible at factor $1/8$, arms will become visible. Appears to be segmented into blocks. For this purpose, the Nearest Neighbor, Bilinear and Bicubic interpolation are used. (`cv2.INTER_CUBIC`) showed that it is able to estimate the missing pixels, but It can't "rebuild" the missing data.
- **A3-A4. Created a stripe object (8 px).** The sampling was done at $\Delta x = 7$ px, aliasing was seen when it is larger than the sampling interval. Nyquist frequency with a false beat period $T_{\text{alias}} = |1/8 - 1/7|^{-1} = 56$ px. Re-quantization with $k = 32, 8, 4, 2$, led to contouring at $k = 8$ and The contour plot has some interesting contours for $k = 4$.
- **A5–A6. Neighborhoods & Distance Measures:** Mapped: N_4, N_D, N_8 define 4-, m-, and 8-adjacency conditions to resolve path ambiguity. For $p = (2, 3)$, $q = (7, 9)$, $\Delta x = 5, \Delta y = 6$: $D_E = \sqrt{5^2 + 6^2} = \sqrt{61} \approx 7.8102$, $D_4 = |\Delta x| + |\Delta y| = 11.0$, $D_8 = \max(|\Delta x|, |\Delta y|) = 6.0$. Verified using NumPy array norms.
- **A7–A8. Affine Transformations & Scaling:** Hand-derived a 2×3 affine matrix for $\theta = 30°, \mathbf{t} = [15, 25]$: $\mathbf{M} \approx [[0.866, -0.5, 15], [0.5, 0.866, 25]]$. Transformed landmark $[50, 80] \to [18.30, 119.28]$, matching `cv2.warpAffine` within 0.5 px. Scaling sweep $s \in [1.0, 1.3]$ verified linear landmark dilation along $\mathbf{x}'(s) = s\mathbf{R}(\theta)\mathbf{x} + \mathbf{t}$.

TABLE I
STEPHAN'S QUINTET GROUND TRUTH (NED / GALAXY ZOO 2)

Galaxy	RA, Dec (J2000)	GZ2 Consensus Class	Dynamical Status
NGC 7317	$22^{\mathrm{h}}35^{\mathrm{m}}52^{\mathrm{s}}, +33°56'42''$	Smooth / Elliptical	Quiescent member
NGC 7318A	$22^{\mathrm{h}}35^{\mathrm{m}}57^{\mathrm{s}}, +33°57'56''$	Distorted / Merger	Active collisional pair
NGC 7318B	$22^{\mathrm{h}}35^{\mathrm{m}}58^{\mathrm{s}}, +33°57'57''$	Distorted / Spiral	Active collisional pair
NGC 7319	$22^{\mathrm{h}}36^{\mathrm{m}}04^{\mathrm{s}}, +33°58'33''$	Barred Spiral	Strongly interacting
NGC 7320	$22^{\mathrm{h}}36^{\mathrm{m}}03^{\mathrm{s}}, +33°56'53''$	Spiral / Clumpy Disc	Foreground spiral

TABLE II
HST VS JWST METRICS & RESTORATION DECISIONS

Instrument	SNR	$\sigma^2_{\nabla^2}$	Dyn Range	Scale
HST ACS (F814W)	28.14	104.22	64.81 dB	$0.050''$/px
JWST NIRCam (F200W)	**85.42**	**312.58**	**92.37** dB	$\mathbf{0.031}''$/px
Pipeline: Wiener deconv → Bilateral ($d = 9$) → Asinh ($a = 0.05$) → RGB				

III. DATASET ACQUISITION & EDA

Validated stack: `astropy` v7.1.0, `astroquery` v0.4.11, `photutils` v2.2.0, `cv2` v4.11.0, `skimage` v0.25.2, `scipy` v1.15.2, `numpy` v2.2.3. Acquired calibrated FITS mosaics via `astroquery.mast`: Stephan's Quintet (JWST NIRCam F090W, F150W, F200W from PID 02732; HST ACS F814W from PID 10602) and Pillars of Creation (HST ACS/WFPC2 from PID 09736). Pixel scales: $0.031''$/px (JWST) vs $0.050''$/px (HST). Processed NaNs via local median imputation, leveled zero-flux offsets, and aligned Galaxy Zoo 2 / NED ground-truth consensus (Table I).

IV. IMAGE RESTORATION & ENHANCEMENT

Week 3 executed a quantitative restoration sprint on Stephan's Quintet:

- **D1. Deconvolution:** Gaussian PSF 7×7 ($\sigma = 1.5$). Sharpness measured via Laplacian variance $\sigma^2_{\nabla^2} = \mathrm{Var}(\nabla^2 I)$. Richardson-Lucy caused ringing at ≥ 20 iters. Unsupervised Wiener deconvolution bounds noise via MMSE inversion without ringing; **Wiener selected.**
- **D2. Spatial Denoising:** Benchmarked Gaussian, Median, Wavelet, NLM, and Bilateral filtering. Bilateral filter ($d = 9, \sigma_c = 75, \sigma_s = 75$) was selected for smoothing flat background noise while strictly preserving sharp galactic edges.
- **D3. Dynamic Range Contrast:** Linear clipped arms; HistEq saturated cores; CLAHE formed halos. Astronomical Asinh stretch (`AsinhStretch(a=0.05)`, 99.5% clip) scales linearly at sky and logarithmically at cores; **selected.**
- **D4. RGB Compositing:** Mapped F200W (R), F150W (G), F090W (B) with per-band `Background2D` (box 50) sky leveling.

V. SOURCE SEGMENTATION & OBJECT ISOLATION

Classical source isolation was executed as follows:

- **E1. Adaptive Thresholding:** Global Otsu severed faint outer discs. Adaptive Gaussian thresholding (`cv2.adaptiveThreshold`, block $= 51$ px, $C = +5$) dynamically preserved diffuse galactic halos.

TABLE III
EXTRACTED CAS MORPHOLOGICAL FEATURES (WEEK 4)

Galaxy	C	A	S	ϵ	e	Reference Class
NGC 7317	1.512	0.000	0.000	0.0774	0.0030	Elliptical (E4)
NGC 7318A	1.842	0.184	0.042	0.2140	0.0232	Interacting Elliptical
NGC 7318B	2.115	0.291	0.088	0.3512	0.0637	Interacting Spiral
NGC 7319	2.450	0.312	0.095	0.4820	0.1239	Barred Spiral
NGC 7320	2.025	0.245	0.112	0.4105	0.0882	Late-type Spiral

- **E2. Morphological Cleanup:** Applied 5×5 elliptical closing (2 iters) → opening (1 iter) → binary hole-filling (`scipy.ndimage.binary_fill_holes`).
- **E3. Star–Galaxy Separation:** Blobs with area $<$ 200 px purged. Evaluated core flux ratio $f_{\text{central}} = \sum_{\text{Core}_{30\%}} I / \sum_{\text{Blob}} I$. Blobs with $f_{\text{central}} > 0.35$ and area < 1500 px purged as stars.
- **E4–E5. Cutouts & Limitations:** Identified and saved five WCS matched galaxy cut-outs (30px margin) and three cut-outs from M16 pillars. Physical problem noted: Contiguous tidal halo blending of NGC 7318A/B.

VI. MORPHOLOGICAL CAS FEATURE EXTRACTION

The classical CAS framework and shape moments were implemented as follows:

- **Concentration (C):** $C = 5 \log_{10}(r_{80}/r_{20})$ from growth curve around flux centroid, enclosing 20% and 80% total light.
- **Asymmetry (A):** $A = \max\left(\frac{\sum |I - I_{180}|}{\sum I} - A_{\text{bg}}, 0\right)$ via 180° rotation with a global-sky-patch noise correction A_{bg}.
- **Smoothness (S):** $S = \max\left(\frac{\sum |I - I_{\text{smooth}}|}{\sum I} - S_{\text{bg}}, 0\right)$ via Gaussian filter ($\sigma = 3.0$ px) minus global sky clumpiness S_{bg}.
- **Geometric Moments:** Binary moments provide area (m_{00}), perimeter, 7 Hu invariants; intensity moments yield centroid, eccentricity (ϵ), and ellipticity (e). Compiled in Table III.

VII. CODE REVIEW FIXES APPLIED

Four problems were solved via an internal code audit: (1) sign correction C = +5 to record weak arms; (2–3) uniformization of A and S to one global sky patch; (4) separation of binary area from flux calculation.

VIII. STATUS & ROADMAP

The entire foundation, acquisition, restoration, segmentation and CAS extraction process is now finished and confirmed. Week 5: introduce deterministic rules-based decision boundaries to CAS space (Elliptical: C 2.2, A ¡ 0.15, S ¡ 0.05; Spiral: C ¡ 2.5, 0.15 A ¡ 0.35, S 0.05; Irregular: A 0.35 or moments distortion) and test against Galaxy Zoo 2 by way of confusion matrix analysis. Week 6: comparison between instruments with HST optical data. Week 7: Final IEEE report and reproducible codebase delivery.